

# Continuous Monitoring Of Minerals In The Soil And To Identify The Diseases Of Banana Plant Based On The Health Of The Soil

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**Abstract—** Agriculture is a strong foundation of economy for many agriculture-based countries like India. Banana is one of the most cultivated agricultural crops especially in African & Asian countries. *Musa Acuminata* is the scientific name of banana. Life span of a banana tree is about 25 years, and it takes around 9-12 months to produce bananas. Production of bananas is affected by various things like natural calamities, diseases, etc.... But diseases are constant threat affecting yield of the crop in agriculture industry, without proper measurements and constant testing it brings a great risk for the production and reduces the quality and the quantity of the cultivation. There are different types of factors causing diseases to banana plants like, viral diseases, health of soil, insects and pesticides etc.... Soil is to be monitored continuously to detect minerals, pH levels etc. to calculate the real-time health of soil. Healthy soil with essential nutrients, water, oxygen, and roots prevents plants from malnutrition diseases and helps plants to flourish and grow healthy and produce good yield.

**Keywords—**Minerals, Insects, Pesticides, PH levels, Diseases

## I. INTRODUCTION

Banana is one of the most essential crop cultivated in Asia & African regions. Numerous factors affects banana farming. Normal catastrophes ex- hurricanes, famine, volcanic eruptions, pollution, acid rains etc.. and

other factors like pest, soil pollution & infertility & insects. In present day farming.



Fig.1. Pictorial representing a Banana Plant (Taken from sqft)

This paper centres around Continuous monitoring of minerals in the soil and to identify the diseases of banana plant based on health of the soil. Monitoring soil health and providing proper minerals and fertilizers improve the banana plant productivity and significantly improves and quality & prevents the plants from malnutrition diseases.

[10] India relies heavily on agriculture, which is a vital component of its economy. The cultivation of bananas serves as a significant source of income in various regions of the country. With approximately 830.5 thousand hectares of land dedicated to banana production, the total output amounts to around 29,779.91 thousand tons. [10]The main states engaged in banana cultivation are Tamil Nadu, Maharashtra, Gujarat, Andhra Pradesh, and Karnataka. However, the changing climate has resulted in an increase in banana plant diseases such as Sigatoka, Cucumber Mosaic

virus, Banana Bacterial Wilt, Bunchy top, and Panama disease.[10]

Traditionally, farmers have relied on expert opinions to diagnose plant diseases, which can be costly and time-consuming. Obtaining expert assistance is particularly challenging in remote areas. Hence, there is a pressing need for an automated system capable of identifying plant diseases through visible indications and providing guidance to farmers to mitigate yield losses.[10] Various techniques for disease identification have been proposed, and among them, the Support Vector Machine (SVM) is a popular classifier for analyzing image data. While SVM is effective for binary classification, real-world problems often require classification into multiple classes. Multi-class patterns have been successfully applied in diverse fields like speech recognition, optical character recognition, bioinformatics, and agriculture.[10]

Generating a multiclass SVM model can be achieved through two approaches: one-versus-rest and one-versus-one. These models combine multiple binary class optimization problems into a single objective function, enabling the classification of multiple disease categories. This study focuses on automating the identification of diseases in banana plants based on visible indications. The system takes banana leaf images as input and classifies the leaves into five possible diseases: Sigatoka, Cucumber Mosaic Virus (CMV), Banana Bacterial Wilt, Bunchy Top, and Panama disease. The research encompasses four stages: image acquisition and pre-processing, image segmentation, feature extraction, and classification.[10]

[10] Farmers have traditionally relied on professional judgments to identify crop illnesses, which can be costly and time-consuming. Obtaining expert assistance in rural areas can be particularly challenging. Therefore, a machine learning system is required to recognize plant illnesses based on their visible symptoms and provide farmers with guidance on minimizing crop losses. Support Vector Machine (SVM), one of the most popular classifiers for analyzing image data, has been proposed as a disease diagnosis method in various research works. [10] While SVM separates data into binary classes, real-world problems often necessitate classification into more than two classes. Multi-class patterns have been employed in different scenarios, including speech recognition, optical character recognition, bioinformatics, and agriculture.[10]

[11] In recent years, the export of bananas from northern Peru has witnessed significant growth, with the Piura region being recognized as a primary banana

export hub by the Ministry of Agriculture and Irrigation (Minagri). Peru's banana exports are predominantly organic, accounting for nearly 3% of global production, according to the FAO. Between 2010 and 2015, the production of organic bananas surged by 94%. This remarkable achievement is attributed to the national organic banana program initiated by Minagri in 1998, which aimed to facilitate the cultivation of "Valery Cavendish" organic bananas in conventional banana soils. Diseases and pests pose significant challenges to banana productivity, with Black Sigatoka being the most devastating disease and the banana weevil causing substantial damage to the banana tops. Banana marchitez has the potential to severely impact national banana production. Therefore, early identification and detection of these disorders are crucial. [11]

The analysis or classification of photographs represents a deep learning application, particularly through Convolutional Neural Networks (CNNs), which are well-suited for processing 2D RGB images. [11] Machine learning applications encompass various areas, including facial recognition, super-resolution imaging, and parameter measurement of goods. In the agricultural industry, RGB images have been employed for pest recognition using machine learning techniques for several years. Hence, a potential solution would be an assistance tool that aids in the identification of banana diseases through automatic capture and analysis of plant leaf images.[11]

## II. ALGORITHM

[8]

The algorithm for temperature, relative humidity, and rainfall data is as follows:

1. The initial input comprises real-time data on temperature, relative humidity, and rainfall. Output the data to the gateway node.
  2. Extract the sensed data from the appropriate sensors and transmit it to the gateway node.
  3. Activate the sleep thread with a 20-minute time interval.
  4. Repeat the loop from step 3. Every 20 minutes, retrieve sensor data for temperature, relative humidity, and rainfall. This time interval is determined based on the requirements of banana crops and agricultural experts. After reading the data, the node enters a sleep mode for the subsequent 20 minutes.
- [8] Rain sensors are utilized to determine rainfall occurrence on the field, which is crucial during the rainy season. These readings provide accurate warnings for Sigatoka disease. The soil moisture algorithm operates similarly, triggering an alert to start the water

pump when the soil moisture falls outside a specific range. [8]

[8] The algorithm for Sigatoka disease alert is as follows:

1. The initial input comprises real-time sensor readings for temperature, relative humidity, and rainfall. Generate an alert for a Sigatoka disease attack.
2. Compare the real-time input values with the threshold values stored in the database.
3. Continuously calculate the total number of weather data over a six-day period.
4. If the cumulative count exceeds 36 within two days, the disease severity is significant; if it exceeds 36 within four days, the severity is moderate; if it exceeds 36 within six days, the severity is normal.
5. Inform the farmer about the disease attack and its severity based on the total number of cases. Using the above-described technique, we alert farmers to the occurrence of Sigatoka disease. This algorithm is continuously executed based on real-time field readings. The cumulative count increases by one when the temperature, relative humidity, and rainfall measurements surpass the threshold values. These counts are accumulated over a six-day period. [8] If the count reaches 36 within two days, it indicates that the farm has experienced two days of favorable weather conditions for Sigatoka disease, such as high or extremely high relative humidity, rainfall, and suitable temperatures. Thanks to the algorithm's continuous and accurate monitoring of the WSN (Wireless Sensor Network), crop losses caused by disease attacks are prevented, and farmers receive timely warnings. [8]

### III. METHODOLOGY

#### A. Classification of soil

The loose material on upper layer of earth formed under specific natural conditions consisting very small particles of minerals/rocks and decayed organic matter which has ability of supporting plants to grow & thrive is known as "soil". Generally soil is composed of organic, inorganic or minerals, air and water. There are different types of soil. They are:

- i. Alluvial Soil
- ii. Black Cotton Soil

- iii. Red & Yellow Soil
- iv. Mountain Soil
- v. Sandy Soil
- vi. Alkaline Soil
- vii. Peaty Soil
- viii. Acidic Soil

#### B. Different Types of Soil

Soil is a valuable natural resource that's plays an important role in sustaining of life on Earth. It's a complex mixture of many minerals, water, air, organic matter, bacteria etc. providing the necessary support for plants to grow & thrive. Soil is not uniform throughout the world, they have remarkable diversity in their composition, texture, and fertility.

##### i) Alluvial Soil :-

They are soils formed by the deposits of surface water. We can find them along banks of riverside, deltas. It is one of the best soil for crop cultivation with highest productivity w.r.t other soils & requires least water due to its high porosity and rich in potassium but poor in nitrogen. Colour of this soil changes from ash grey to light grey.



Fig 2. Sample of Alluvial Soil

##### ii) Black Cotton Soil :-

Colour of black cotton soil differs from grey to deep black. It is generally clayey, deep & impermeable. This soil is extremely fertile. It is rich in elements like potassium, aluminium, iron, lime and magnesium but it is deficient in nitrogen, phosphorus and organic matter and it have great shrinkage & swelling capacity and well known for its moisture holding capacity.



Fig 3. Sample of Black Cotton Soil

iii) *Red & Yellow Soil:-*

Red & Yellow soil is mostly found in areas having very less rainfall. It is in reddish yellow colour. The reddish colour is because of the presence of iron element in metamorphic and crystalline rocks. When this soil is in hydrated form it appears yellow. Generally such types of soils are deficient in humus and elements like phosphorous and nitrogen.



Fig 4. Picture of Sample Red & Yellow Soil



Fig 5. Field of Red & Yellow Soil

iv) *Mountain Soil:-*

It is found in the valley and sloppy regions of the mountains at higher altitudes and valleys. Generally formed by deposition of organic matter and heterogeneous in nature based on parent rocks, climate and ground configuration. The properties varies with mountainous climate and altitude. These are deficient in potash, phosphorus but high amount of humus.

v) *Sandy Soil:-*

It is mostly found in desert areas or in dry areas with very less to no rainfall regions and at the banks of sea and oceans. The salt content is so high in this soil which makes the nature of sandy soil saline. As these soil has quick drainage of water most of the crops are not suitable to grown in this soil.



Fig 6. Field of Sandy Soil

vi) *Alkaline Soil:-*

Alkali or alkaline soils have poor infiltration capacity & poor structure of soil with high pH (>8). These contains high amount of calcium, sodium, magnesium. These are clay soils. The name of the soil is derived from the alkali metal group of elements where the sodium belongs and which can induce basicity. These soils are also referred as alkaline sodic soils.





Fig 7. Sample of Alkaline Soil

vii) Peaty Soil:-

This type of soil is formed by the accumulated decomposition of organic materials under waterlogged environment where there is lack of air supply or oxygen. It has large porosity character, low density and large water & organic matter and nutrient deficient.

viii) Acidic Soil:-

It is mostly found in humid regions with heavy rainfall. Most of acidic soils are found in hilly terrains. These soils are sour & acidic in nature and have pH value less than 6.5. Such type of soils are generally have deficiency in nutrients.



Fig 8. Sample of Acidic Soil

#### IV. TYPES OF DISEASES FOR BANANA PLANT

Bananas are highly susceptible to a range of diseases caused by fungi, bacteria, and viruses. These diseases

can have significant effect on the yield and quality of the banana crops, leading to reduced incomes for farmers and higher prices for consumers. Therefore, it is essential to monitor and manage the health of banana plants to ensure a healthy harvest.

It is caused by a type of fungus called *Fusarium oxysporum f.sp.cubense* and is one of the most destructive diseases of banana crops. The fungus infects the roots of the plant, causing them to rot and preventing the uptake of water and nutrients. This leads to wilting, yellowing, and ultimately death of the plant. Panama disease can be spread through contaminated soil, water, and plant material, making it challenging to control. However, by using good sanitation practices, such as removing and destroying infected plant material, farmers can reduce spread of diseases. Additionally, monitoring the nitrogen and phosphorus levels in the soil can help prevent Panama disease, as these minerals have been shown to reduce the severity of the disease.

[14] Now the culture is endangered by different diseases. At the moment, the use of pesticides is more efficient, but increasing the use of pesticides can treat the environment. To overcome this we can there is another way by using biological control using natural enemies of these pests and it is observed that *Bacillus* species are more effective role of biogents.

Microbial inoculants and organic fertilizers have been identified for productivity and Growing plants; it is recognized worldwide as an alternative source of fertiliser.[14] Rhizobacteria suitable for plant growth (PGPR) are used to improve nutrients and other sources of fertiliser. PGPR can improve bacterial colonization at the root surface and places where there are more microbial cells were present.[14]

The diseases of banana plant are as follows as – Black Sigatoka, Panama Wilt / Banana Wilt Disease(BXW), Banana Bunchy Top(BBTV), Rhizome rot, Anthracnose, Moko disease, Tip Over, BBMV - Banana bract mosaic virus, BSV - Banana streak disease, CMV - Infectious chlorosis.

##### i. Black Sigatoka, *Mycosphaerella* leaf spot:

**Black Sigatoka** is another significant threat to banana crops worldwide. It is caused by the fungus *Mycosphaerella fijiensis*, which infects the leaves of the plant, leading to premature death of the leaf. This reduces the photosynthetic capacity of the plant, which in turn reduces yield. Black Sigatoka can be managed through the use of fungicides, planting disease-resistant cultivars, and maintaining proper nutrition levels in the

soil. By monitoring soil minerals such as nitrogen, phosphorus, and potassium, farmers can ensure that their banana plants have the necessary nutrients to resist the disease.

## ii. Bunchy Top (or) Curly Top:

**Banana Bunchy Top Virus (BBTV)** is a viral disease that affects the entire banana plant. The virus causes stunted growth, yellowing of the leaves, and a bunchy appearance at the top of the plant. BBTV is primarily spread by the banana aphid, which feeds on infected plants and then spreads the virus to healthy plants. There is no cure for BBTV, and infected plants must be removed and destroyed to stop the spread of diseases. However, farmers can reduce the danger of BBTV by using virus-free planting material, planting disease-resistant cultivars, and controlling the banana aphid population through the use of insecticides.

## iii. Panama Wilt(or) Banana Wilt Disease(BXW):

**Banana Wilt Disease (BXW)** is caused by the bacterium *Xanthomonas campestris* and is characterized by wilting and death of the entire banana plant. The bacteria can be transmitted through contaminated soil, water, and plant material, and can persist in soil for several years. There is no known cure for BXW, and infected plants must be removed and destroyed to stop the wide spread of the disease. Monitoring levels of the potassium in the soil can help prevent BXW, as potassium has been shown to reduce the severity of the disease.

## iv. Rhizome Rot:

The Symptoms of **Rhizome Rot: *Erwinia chrysanthemum*** disease are as follows - This type of disease is more notable on young suckers which was leading to rotting and emitting of foul order, the breaking of pseudostem is familiar in later stages of infection in cultivars robusta grand nine, In beginning stage of the infection, the dark brown or yellow water soaked areas are more in cortex area.

The management of Rhizome Rot will be as follows – Rotation of crop with sugarcane or rice should be for 3 to 4 years, you must ensure that only planted dip suckers or healthy suckers in copper oxychloride,

(0.5g/lr) + monocrotophos 2.5ml/lr solution

## v. Anthracnose:

In the early stages small, round, black spots appears on the affected fruit. At that point these spots broaden in estimate, turn to brown colour.

The skin of natural products darkens, fades and becomes covered with a characteristic pink acerbilin. Eventually, the entire finger is affected. In the future, the disease spreads and affects entire groups.

This disease causes premature ripening and wrinkling of fruits that are covered in pink spore masses. If black spots develop on the stem, then the stem will wither and your fingers will fall off your hand. Sometimes the main stem of a bunch may get sick then the infected fruits turns to black color and rot.

The spread of the disease is carried out by airborne conidia, as well as numerous insects that stay at banana flowers at frequent times.

The disease is favourable by high temperature and humidity, cuts and bruises on the fruit, and the susceptibility of the cultivar.

This disease gets appeared whenever the deficiency of the minerals get reduced.

## vi. Moko disease :

The Moko Disease is also known as Bacterial Wilt.

This Disease Symptoms are as follows - The leaves gets turned into yellow and move up. The petiole is broken and the leaves droop. At the time of incision, there is a discoloration ranging from light yellow to dark brown in the blood vessel area. The discolouration will be appeared in the central part of the corm. The inside of the fruit rots and turns dark brown.

Whenever the pseudostem has been cut transversely then the bacterial ooze can be seen clearly.

The bacteria will survive on infected plant material, vegetative reproductive organs, wild host plants, and the soil.

Common favorable conditions for the disease are high temperatures and high soil moisture.

## vii. Tip Over:

Tip Over is also known as bacterial soft rot. This disease is more pronounced on young shoots, resulting in rot and an unpleasant odour.

If the affected plants are peeled out, then it comes out from the collar region and then it leaves the corm with their roots in the soil.

In the early stages of infection, dark brown or yellow watery areas are more common on the bark. When an

affected plant is cut, then yellow or red silt is visible on the neck. Bacteria on the plant will survive on the plant debris and will infect by watering through damaged tissue. It is worse in wet hot weather and the bacteria will get spreaded in polluted water. Higher temperatures as well as high humidity are the ideal conditions for the bacteria.

#### **viii. Banana bract mosaic virus (BBMV):**

The disease symptoms are as follows - The illness is identified by pink to ruddy spindle-shaped groups on the pseudostem, midrib, and on the peduncle.

Typical mosaic and spindle-like light mosaic stripes on the bracts, stems and on the fingers were also detected. Suckers display an unusual reddish-brown stripe as they emerge and detach the leaf sheath from the central axis. Characteristic symptoms are travel palms, elongated flower stalks, and crown leaf accumulation with the appearance of a half-double hand. The Survival and the spread of this disease are as follows - This illness is caused by a infection that has a place to the potyvirus bunch. Virions are flexible and filamentous. The virus is transmitted by aphid vectors such as *Aphis gossypii*, *Pentolonia nigronervosa*, and *Rhopalosiphum maidis*. Infection are mainly spreaded by runners outdoors.

#### **ix. Banana streak disease (BSV):**

The Disease Symptoms of this disease are as follows - A prominent symptom of BSV is the appearance of yellow streaks on the leaves, which gradually dies and black streaks appear on older leaves.

The Survival and the spread of this disease are as follows - The virus is primarily transmitted through infected plant material, but is also believed to be transmitted by the mealybug (*Planococcus citri*) and, more likely, *Saccharicoccus sacchari*.

In shoot tip culture, shoot tips cannot be removed from vegetatively propagated material.

#### **x. Infectious chlorosis (CMV):**

The Symptoms of this disease are as follows - In all the stages of the growth of crop, this disease shows itself.

Due to repeated use of injections of infected plants, the disease spreads, leading to a decrease in yield and quality.

This type of disease may occur in all the states of banana growth. There are pale yellow stripes parallel to the veins, and the stripes are visible on the leaves.

The streaks usually runs between mid rib and edge of the blade. This type of disease is distributed by suckers and *Aphis gossypii*.

**Source :** <https://vikaspedia.in/agriculture/crop-production/integrated-pest-managment/ipm-for-fruit-crops/ipm-strategies-for-banana/diseases-and-symptoms>

#### **Ripening of banana:**

[1] Ripening is one of the most important phase in agriculture. Ripening of fruits are categorized into two types either climacteric or non-climacteric. Non-climacteric fruits do not ripen off the tree after harvesting whereas climacteric fruits ripen even after harvesting. Banana belongs to climacteric fruit. Generally, bananas are green in colour and slowly turns into yellow after ripening. Bananas are ripened in chambers with controlled temperature of around 15-18 degrees, humidity, and ethylene gas at 100-150 ppm concentration with exposure time of 24 to 48 hours.[1] The peel colour of banana transforms from green to yellow colour. When the banana peels is converted from green to yellow then the starch is converted into sugars and pulp gets soft. Bananas are well cleaned, pre-cooled and well processed at full yellow colour and exported into the market. [1]

Improper temperature and insufficient amount of ethylene gas leads into problems like chilling injury & unripen bananas. It can be well noticed by observing its peel colour. Colour of banana peel is an important key to maintain the level of temperature and ethylene in ripening process. An expert operator's attention is required in this entire banana ripening process to get successful results.[1]

[1] There are also some automated process for this ripening which sorts, controls and adjusts the temperature, humidity & concentration of ethylene gas by observing the colour changes of the banana peel. [1] Electrical methods, Spectral methods & Image processing Methods are some of the processes used to detect the level of ripeness of the fruits in this ripening process. These methods prevent and warns us if any abnormal activity occur in the process. [1]



Fig.9 Stages of Ripening of banana fruit.

Reference : Irzal Ahmad Sabilla, Cahyaningtyas Sekar Wahyuni, Chastine Fatichah and Darlis Herumurti, "Determining Banana Types and Ripeness from Image using Machine Learning Methods", 2019.

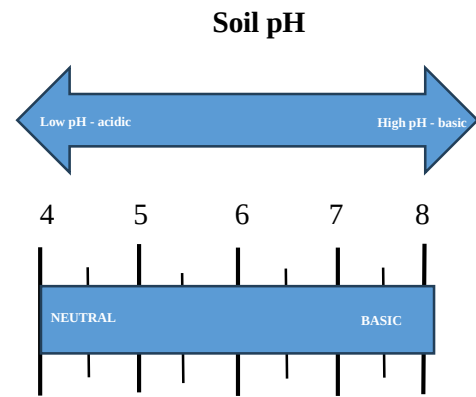


Fig 10. pH Value Ranges

### Type of earthworm in the banana tree:

[2]Earthworms dwell in the soil and undergrounds of the earth. Generally, people use them as bait for fishing, making medicine and in agriculture. Earthworms are found in the soil around banana trees or crops, because the soil around the banana trees has plenty of moisture and with low temperature which is preferred by earthworms. Presence of earthworms benefits a lot for the banana tree habitat. Earthworms have both environmental chemical and physical requirements with optimum humidity, high organic matter etc..., [2]

The physical and chemical factors of life of earthworms are pH, moisture, temperature, soil type & structure, humidity, organic matter, porosity of the soil and limiting factors are elements of carbon and nitrogen. Existence of earthworms makes the soil more fertile and increases its porosity which helps in flowing of air into the soil and also helps water to penetrate deep into the soil and helps the tree roots to get required amount of water. Earthworms play an important role in soil ecology as they provide plenty of organic matter through their stool and also increase the porosity of the soil.[2]

### V. SOIL PH

It is one of the most important factor in growth of the plant as it influences the nature of the soil, availability of nutrients, toxicity & the activity of organisms in the soil.

#### A. pH Ranges

- < 4.6 Extreme acidic
- 4.6 – 5.5 Strong acidic
- 5.6 - 6.5 – moderately acidic
- 6.6 – 6.9 slightly acidic
- 7.0 – neutral
- 7.1 – 8.5 Moderately alkaline
- > 8.5 – strongly alkaline

### IV. MICRO NUTRIENTS AND IT'S DEFFICIENCY SYMPTOMS

For plants Soil is a major source of nutrients for growth. Nitrogen(N), phosphorus(P) and potassium(K) are the three main nutrients, they together trio known as NPK.

#### A. Nitrogen

It's a key nutrient in plant growth. Its found in all plant cells, proteins, hormones, chlorophyll etc.. Atmospheric nitrogen is a source of soil nitrogen. Soils having high organic matter are generally higher in nitrogen. Photosynthesis occurs at high rates when there is sufficient nitrogen.

#### Deficiency symptoms :-

- i. Stunted growth
- ii. Lower protein content
- iii. Flowering is greatly reduced
- iv. Leaves turning into light green to pale-yellow colour from tips followed by shedding/death based on degree of deficiency
- v. Leaching
- vi. Consumption by plants and other organisms



## B. Phosphorus

It helps in transferring the sunlight energy to all parts of the plants and helps in plant growth & stimulating early root growth. All manure contains phosphorus, and its rich in source in manure from grain-fed animals. Generally concentration of phosphorus in plants ranges from 0.1% to 0.5%.

### Deficiency Symptoms:-

- Overall stunted appearance
- Root development restriction
- Poor fruit & seed development
- Delaying of maturity
- Matured leaves have dark to blue-green colour characteristics.

## C. Potassium:-

Potassium helps by improving disease resistance of plans, helps in moving and formation of sugars, starches and oils and also improves the quality of fruits. It involves in many enzymatic reactions, involves in helping during transpiration.

### Deficiency Symptoms:-

- Slow and stunted growth
- Shrivelled seeds of fruits
- Stalks weaken and plant lodge easily

### D. Some common deficiency symptoms are:-

Fall of pre-matured buds & leaves – deficiency of K&P. Retarded plant growth – deficiency of N, P, K. For banana plant, pH value should be around 5.5-6.5, N:k ratio should be 1:1 or 1:1.6 depending on method and yields.

|                              |                    |
|------------------------------|--------------------|
| Phosphorous (P)              | 80 ppm             |
| Potassium                    | 0.5 meq /100g      |
| Calcium (Ca)                 | 4 to 10 meq / 100g |
| Magnesium (Mg)               | 1 to 3 meq / 100g  |
| Electrical conductivity (EC) | <0.15 ds/m         |

TABLE I. GUIDE TO PREFERRED LEVELS FOR BANANAS FROM A TYPICAL SOIL TEST

Source: <https://www.haifa-group.com/banana-fertilizer/crop-guide-banana-fertilizer-recommendations>

The order of nutrient uptake and removal uptake of macronutrients is –

(K)Potassium > (N)Nitrogen > (Ca)Calcium > (Mg)Magnesium > (P)Phosphorus.

Young banana plants needs around 0.25 to 0.5 pounds of fertilizers per month. A balanced fertilizer of 8-10-8 (NPK) is recommended. Generally bananas needs an average of 4 to 6 inches of water every month, or about 1 – 1.5inch every week, depending on the season. Ensure that the soil drains well & doesn't have standing water as overwatering leads to root rot.

## V. COMPARATIVE ANALYSIS

At initial stage of planting the banana plant of acidic soil, the values will be as follows:

| Type                  | Value            |
|-----------------------|------------------|
| PH                    | 4.90             |
| Organic Carbon        | 0.76%            |
| Available Nitrogen    | 389.01 kg/hector |
| Available Phosphorous | 21.30 kg/hector  |
| Available Potassium   | 452.79 kg/hector |
| Available Sulphur     | 27.32 kg/hector  |
| Zinc                  | 0.63 bpm         |
| Boron                 | 3.45 bpm         |
| Iron                  | 26.47 bpm        |
| Manganese             | 10.34 bpm        |
| Copper                | 0.58 bpm         |

Table2: Values at initial stage of banana plant of acidic soil

From the above Table2, The Above values are calculated based on the temperature and these are calculated at initial stage of planting.

After a 15 days of testing, the values may be increased or decreased. If the values are decreased then fertilizers will be added in the soil to maintain the soil fertility. Then the values will be increased.

At Fertilization stage the values will be as follows:

| Type                  | Value            |
|-----------------------|------------------|
| PH                    | 4.96             |
| Organic Carbon        | 0.81%            |
| Available Nitrogen    | 391.01 kg/hector |
| Available Phosphorous | 21.34 kg/hector  |
| Available Potassium   | 453.7 1kg/hector |
| Available Sulphur     | 27.38 kg/hector  |
| Zinc                  | 0.69 bpm         |
| Boron                 | 3.50 bpm         |
| Iron                  | 26.51 bpm        |
| Manganese             | 10.38 bpm        |
| Copper                | 0.59 bpm         |

Table3: Values at fertilization stage of acidic soil

From the above Table3, the above values are the values at fertilization stage. Compared to initial stage and fertilization stage. In initial stage, the values are getting decreased. In fertilization stage, the values are increased by adding fertilizers in the soil.

In the above two tables i.e, table-2 & table-3, PH value has been increased with 0.06% because at fertilization stage the fertilizers are added to increase the value. Similarly the values of available nitrogen, Organic carbon, available potassium, available phosphorous, available boron, zinc, sulphur, Iron, Manganese, Copper values have been increased due to the addition of fertilizers in the crop to get best and accurate values.

As per our knowledge, here's the comparative analysis of the minerals in the soil. At the initial stage of planting the banana plant of acidic soil the values which we have got are not efficient and does not support the crop to grow. So to resolve this problem, the fertilizers will be added at later stage. Whenever the fertilizers is been added, then the values will be increased and the values we get are efficient when compared to initial stage. And then the diseases will be identified based on the values which we have got while testing the soil and the diseases can also be identified based on the health of the soil.

## CONCLUSION

The monitoring soil minerals is crucial for preventing and identifying diseases that affect banana crops. By ensuring that the soil has the necessary nutrients, farmers can promote healthy growth and yield of their banana plants. Additionally, good sanitation practices, regular inspection of banana plants for signs of disease, and the use of disease-resistant cultivars can further reduce the risk of disease. By taking these steps, farmers can protect their crops, reduce losses, and provide consumers with high-quality bananas.

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